ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

DATABASE MANAGEMENT SYSTEM

ASSIGNMENT 4

RELATIONAL DATABASE VS. NOSQL DATABASE

Hotel Management System

Particular Details	**Information**
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Academic Year	2nd Year

Introduction

Relational and NoSQL databases are used to store and manage application data, but they use different data models. Relational databases organize information into tables and use relationships between tables. NoSQL databases support models such as documents, key-value pairs, columns, and graphs. This assignment compares both approaches and considers the Hotel Management System.

1. Comparison Between Relational and NoSQL Databases

Feature	**Relational Database**	**NoSQL Database**
Data structure	Tables with rows and columns.	Documents, key-value pairs, columns, or graphs.
Schema	Usually has a predefined structure.	Often supports flexible structures, depending on the database.
Relationships	Uses primary keys, foreign keys, and joins.	May use embedded documents, references, or model-specific links.
Consistency	Commonly supports ACID transactions and strong consistency.	Consistency guarantees depend on the product and configuration.
Scalability	Often scales vertically; horizontal scaling is also possible.	Many systems are designed for horizontal scaling.
Querying	SQL supports joins, filtering, grouping, and aggregation.	Query methods depend on the NoSQL database and its data model.
Data integrity	Supports constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK.	Integrity features vary by database and application design.
Examples	MySQL, PostgreSQL, Oracle Database.	MongoDB, Redis, Apache Cassandra.

2. Why Organizations Choose Different Database Technologies

Relational Databases

- Suitable for structured information with clear relationships.
- Keys and constraints help maintain data accuracy.
- Transactions help keep related operations consistent.
- SQL makes it possible to join and summarize information from multiple tables.
- Commonly used for reservations, payments, inventory, banking, and management systems.

NoSQL Databases

- Useful when information has flexible or changing structures.
- Some products are designed to distribute large workloads across multiple servers.
- Can suit document catalogues, caching, content management, and particular real-time applications.
- The best choice depends on the required data model, consistency, workload, and operating needs.

3. Comparative Report with Real-World Examples

Application	Suitable Type	Example	Reason
Hotel Management System	Relational	MySQL	Connects guests, rooms, reservations, services, and payments through keys.
Banking system	Relational	PostgreSQL	Structured accounts and transactions need reliable transaction handling.
Flexible product catalogue	NoSQL	MongoDB	Documents can store different attributes for different product types.
Distributed caching	NoSQL	Redis	Key-value storage supports fast access patterns.
Large distributed data workload	NoSQL	Apache Cassandra	Designed for distributed storage and horizontal scaling.

4. Application to the Hotel Management System

A relational database such as MySQL is a suitable choice for the proposed Hotel Management System because the application contains structured information and clear relationships.

- Guest records can be connected to reservations using guest_id.
- Rooms can be connected to room categories using room_type_id.
- Reservations can reference rooms and guests using foreign keys.
- Payments and service usage can be linked to reservations.
- SQL joins can generate reports showing guest names, room numbers, reservation dates, and payment details.

A NoSQL database could be considered later if the system needed flexible document storage or large-scale distributed workloads. For the current design, the relational model is a straightforward and practical choice.

Conclusion

Relational and NoSQL databases have different strengths. Relational databases are well suited to structured records, integrity constraints, and relationships, while NoSQL databases can suit flexible data models and certain distributed workloads. MySQL is appropriate for this Hotel Management System because it supports the required relationships between guests, rooms, reservations, services, and payments.